

FIN ST402–ST416

Independent Orbit Persistence, First Forced Invariant Relations, and a Certified Limiting Infrared Face

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Abstract

Programs ST402–ST416 continue the local analytic and computational FIN research program. The principal result is an independent reconstruction of the exact twelve-minimum theorem for the supplied dimensionless strict functional at gain $g = 4$. Unlike the preceding proof, the replay does not read the earlier cap or root certificates. It rebuilds the strict matrix, a rational feasible benchmark, a 10,000-cell outward exclusion, a 121-pair curvature reduction, and an independent Krawczyk root enclosure. The proof can in fact be simplified: compactness, outside-cap separation, strict cap convexity, and transitive D_{12} symmetry alone force exactly twelve global minima in one orbit. The same conclusion is certified uniformly for every $g \in [3.999, 4.001]$.

Additional results prove an abstract orbit-exhaustion theorem with axiom-removal counterexamples, the exact order-two stabilizer, a conditional local gradient-flow basin, at least 136 forced invariant relations in degree eight, and a sharp adapted-metric Chernoff contraction gap that improves the former conservative bound by more than 69 decimal orders. A continuum-IR Krawczyk parameter tube is enlarged by a factor 55, and a compactified upper-band-loss face is isolated at $T \approx 2.4421819771$. The attachment of the finite branch to that face remains numerical. No result derives the gain, selects an orbit member, supplies dimensional units, completes the legacy-to-strict bridge, or provides independent empirical evidence.

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1 Scope and epistemic boundary

The audited repository baseline is Git commit `bb540a767e210a5a3bf816e4868e7bcefbac3872`, supplemented by the local ST387–ST401 packet and the executable artifacts of this release. The strict working operator is the finite circulant Laplacian

$$A = sI - W, \quad W_{xx} = 0, \quad W_{xy} = \frac{\cos(0.18575 d(x, y) + 0.16250)}{1 + d(x, y)^{9/5}} \quad (x \neq y), \quad (1)$$

on $\mathbb{Z}_{12}$. Its off-diagonal row sum is $s = 1.660307278766099 \dots$. The central functional is

$$V_g(p) = D(p||u) - \frac{g}{2}(p - u)^T A(p - u), \quad D(p||u) = \sum_{i=0}^{11} p_i \log(12p_i), \quad u_i = \frac{1}{12}. \quad (2)$$

All quantities in this release are dimensionless.

The legacy ontological kernel remains an intermediate bridge object. No legacy electroweak, electromagnetic, gravitational, or dimensional role is transferred to (1). The repository ontology—the nadsoliton itself as primordial fractal information in a solitonic state—is retained as project context rather than inserted into the finite proofs.

The main theorem is deliberately narrower than a physical symmetry-breaking claim. An orbit of twelve degenerate minima does not choose one member. Thus the exact result strengthens, rather than removes, the selector question: QW-2191 remains open. Likewise, the theorem states what follows from a supplied g ; it does not generate that gain.

2 Executive result ledger

Program	Status	Result
ST402	[PROVEN]	Independent interval replay: exactly twelve global minima of V_4 , one D_{12} orbit.
ST403	[PROVEN]	Abstract finite orbit-exhaustion theorem and five axiom-removal counterexamples.
ST404	[PROVEN]	The stabilizer of a minimum has order two and is exactly a vertex reflection; orbit size is twelve.
ST405	[PROVEN]	The twelve-minimum theorem holds uniformly for every $g \in [3.999, 4.001]$.
ST406	[PARTIAL]	At least one global orbit change must occur, but the complete gain-transition diagram and uniqueness remain open.
ST407	[PROVEN] core / [STRONG NUMERICAL EVIDENCE] catalog	Uniform and localized minimum orbits are certified; nine sampled stationary orbits form a nonexhaustive Morse catalog.
ST408	[PROVEN] conditional	A declared projected gradient flow has an explicit invariant local basin and certified exponential rate.
ST409	[BLOCKED]	No new strict typed source of the signed nonzero gain was found.
ST410	[PROVEN] method no-go	Both the positive-weight envelope and strict-cap transfer fail for the rank-seven mediator; globality remains open.
ST411	[PROVEN]	Rank-nullity forces at least 136 degree-eight relations in the declared even invariant-generator family.
ST412	[PROVEN] algebraic / [STRONG NUMERICAL EVIDENCE] numerical	Exact coefficient rank 365 coexists with condition number about 1.50×10^4 under a supplied finite design.

Program	Status	Result
ST413	[PROVEN] conditional	An adapted metric raises the explicit synthetic Chernoff gap from $10^{-71.60}$ to 1.99×10^{-2} .
ST414	[PROVEN] conditional	A common Krawczyk box certifies the continuum-IR root over a parameter rectangle 55 times wider than ST395.
ST415	[PROVEN] limiting face / [STRONG NUMERICAL EVIDENCE] attachment	A unique compactified upper-band-loss face is interval-certified; branch attachment remains numerical.
ST416	[BLOCKED]	Strict source, physical scale, laboratory record, and external-evidence gates remain open.

3 ST402: independent reconstruction of the global orbit theorem

3.1 Independence of implementation

ST402 reconstructs the transcendental matrix (1) directly from its four declared constants. It does not import the ST389/ST390 cap code, their root packet, or their conclusion. A qualitative radial seed is solved anew, rounded to an exactly normalized rational probability with denominator 10^{18} , and evaluated by outward interval arithmetic. The resulting feasible value is approximately

$$V_4(p_{\text{rat}}) = -0.845198507679593 \dots$$

The independently built radial stationarity system is also enclosed by an eight-dimensional Krawczyk box of radius 10^{-8} ; its minimum component margin is 9.9866×10^{-9} . This root enclosure is a reproducibility cross-check, but the simplified global proof below does not require an inherited root.

3.2 Outside-cap exclusion

Let $r = \|p - u\|_2^2$. If $\max_i p_i \leq m_0 = 0.94$, then $r \leq m_0 - 1/12$. The entropy minimum at fixed collision is

$$D_{\min}(r) = a \log(12a) + 11b \log(12b), \quad a = \frac{1 + \sqrt{132r}}{12}, \quad b = \frac{1 - a}{11}. \quad (3)$$

Positivity of the strict off-diagonal weights gives the energy upper bound

$$(p - u)^T A(p - u) \leq (s + w_6)r + \frac{s}{12} - \frac{11w_6}{12},$$

where w_6 is the smallest row weight. A complete 10,000-cell outward cover of (3) minus this energy bound gives

$$\inf_{\max p_i \leq 0.94} V_4(p) - V_4(p_{\text{rat}}) \geq 9.5520838498 \times 10^{-5} > 0.$$

The origin cell is paid separately by $D \geq 0$ and $\lambda_{\max}(A) \leq 2s$. Therefore every global minimizer belongs to one of the twelve disjoint caps

$$\mathcal{C}_i = \{p \in \Delta_{11} : p_i > 0.94\}.$$

No simplex-boundary point minimizes (2): moving an amount ϵ into any zero coordinate contributes $\epsilon \log \epsilon$, which dominates the finite directional derivative of the quadratic term.

3.3 Strict cap convexity

In the interior, the tangent Hessian is

$$\nabla^2 V_g(p) = \text{diag}(p_i^{-1}) - gA.$$

On $\mathcal{C}_0$, write $a = -v_0$ and normalize the ℓ^1 mass of the remaining components to one. Cauchy's inequality gives

$$\sum_i \frac{v_i^2}{p_i} \geq a^2 + \frac{1}{1 - m_0}.$$

At fixed positive and negative masses, convexity of $v^T Av$ places its maximum at a pair of vertices. Thus only $11^2 = 121$ one-variable interval quadratics must be checked. The worst outward result is

$$\sum_i \frac{v_i^2}{p_i} - 4v^T Av \geq 0.624323055377694,$$

or 0.312161527688847 in the recorded Euclidean normalization. Every cap is strictly convex.

Theorem 1 (Independent strict orbit theorem). *For the supplied strict operator (1) and supplied gain $g = 4$, the set of global minimizers of (2) consists of exactly twelve points and is one D_{12} orbit.*

Proof. Continuity on the compact simplex gives a minimizer. The outside cover puts it in one cap. Transitivity of the invariant D_{12} action moves it into every cap, giving at least twelve distinct minima because the caps are disjoint. Strict cap convexity permits at most one minimum per cap. Hence there are exactly twelve. $\square$

4 ST403–ST404: abstraction, necessity, and stabilizer

The preceding proof exposes a reusable theorem.

Theorem 2 (Finite orbit exhaustion). *Let a finite group act transitively on the n coordinates of a simplex and let F be continuous and invariant. Suppose $m > 1/2$, one feasible value is strictly below every value on $\{\max_i p_i \leq m\}$, and F is strictly convex on each cap $\{p_i > m\}$. Then $\operatorname{argmin} F$ has exactly n points and is one group orbit.*

Each hypothesis is operationally necessary for this proof architecture. Removing compact existence allows $F(x) = x$ on $(0, 1)$; removing separation allows the uniform point to win; removing cap convexity allows flat continua; removing transitive invariance allows unequal cap depths; and allowing $m \leq 1/2$ destroys the disjoint cap count. These counterexamples do not prove that no alternative theorem is possible; they prove that no listed premise is decorative here.

The cap-zero minimum has a unique maximum at vertex zero. Uniqueness inside that cap forces invariance under the reflection $j \mapsto -j$. Any D_{12} stabilizer must fix the unique maximum, and exactly two group elements fix vertex zero: the identity and that reflection. Hence

$$|\operatorname{Stab}(p_*)| = 2, \quad |D_{12} \cdot p_*| = \frac{24}{2} = 12.$$

This proves the degeneracy exactly; it does not break it.

5 ST405–ST408: robustness, transition limits, and declared dynamics

5.1 A uniform gain tube

For the fixed rational benchmark, the difference between its quadratic energy and the outside energy envelope stays positive throughout the collision interval. Consequently outside separation is worst at the lower gain endpoint, whereas cap curvature is worst at the upper endpoint. Paying both endpoints yields

$$g \in [3.999, 4.001] : \quad \Delta_{\text{outside}} \geq 6.23664718 \times 10^{-5}, \quad \mu_{\text{cap}} \geq 0.3100312347.$$

Corollary 1 (Uniform persistence). *For every supplied $g \in [3.999, 4.001]$, V_g has exactly twelve global minimizers forming one D_{12} orbit.*

This is a nonzero robustness theorem. It does not source any point in the gain interval.

5.2 What is and is not known about the transition

At $g = 0$, relative entropy has the unique global minimizer u . At every gain in the ST405 interval, the global minimizer is nonuniform. Therefore at least one global minimizer-orbit change occurs in $(0, 3.999]$. The uniform Hessian loses local positivity only at the numerically evaluated value

$$g_{\text{Hess}} = \frac{12}{\lambda_{\max}(A)} = 5.1234275514\dots$$

Earlier interval work isolated equality between the uniform value and one localized stationary branch in

$$[2.902496471747767, 2.902496491747767].$$

That is a certified *branch-local* coexistence bracket, not a global-transition theorem. ST406 therefore rejects the requested stronger classification rather than renaming local coexistence as a global or physical critical point.

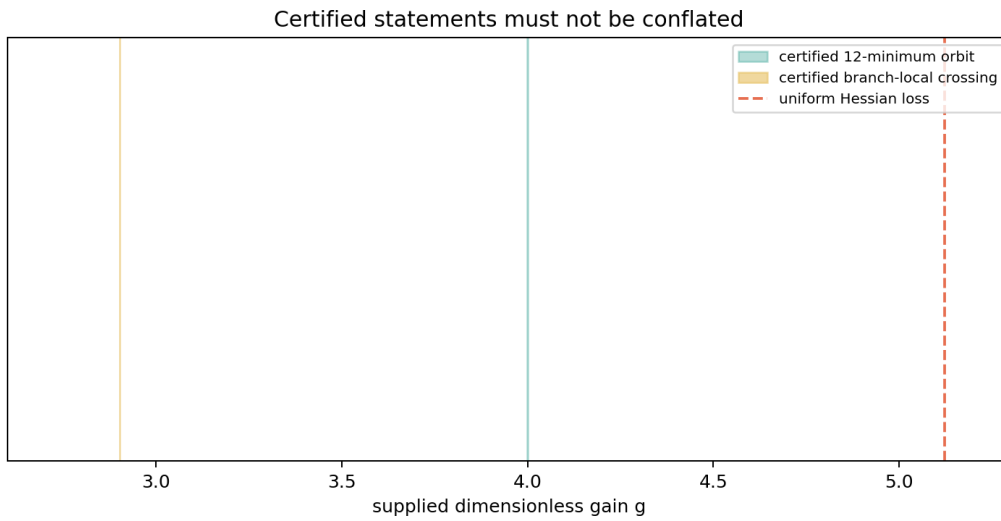


Figure 1: Three different statements: branch coexistence, certified global persistence, and uniform Hessian loss.

5.3 Morse landscape and local flow

The uniform stationary point is an exact local minimum at $g = 4$, with minimum tangent Hessian eigenvalue $2.63127\dots$. The twelve localized global minima have Morse index zero by the cap certificate. In addition, 1,200 deterministic starts found nine distinct D_{12} stationary orbits: two numerical index-zero orbits, four index-one orbits, and three index-two orbits. Only the uniform and localized minimum orbits are certified. The saddle catalog is not an interval exhaustion.

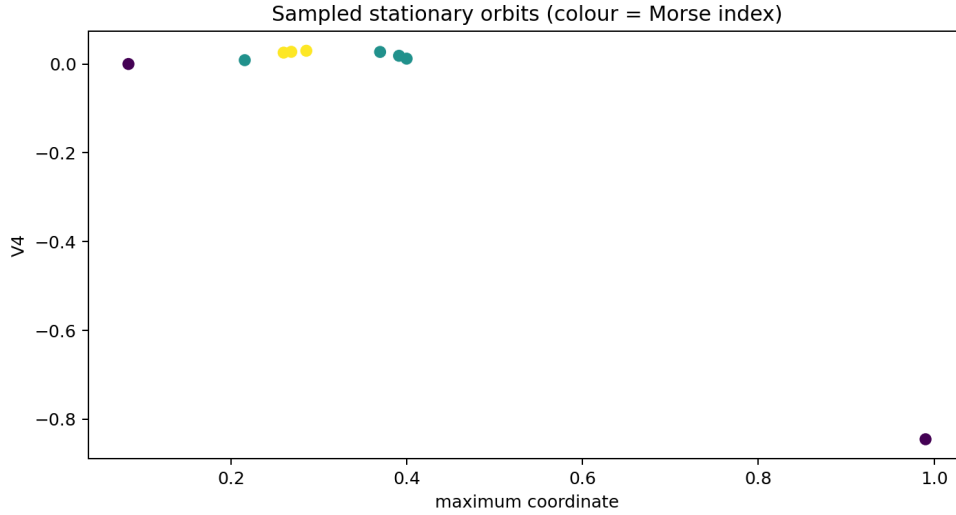


Figure 2: Nonexhaustive sampled stationary-orbit catalog. Colour encodes sampled Morse index.

For the explicitly supplied projected Euclidean flow

$$\dot{p} = -P_T \nabla V_4(p), \quad P_T = I - \frac{\mathbf{1}\mathbf{1}^T}{12},$$

the independent Krawczyk enclosure and cap curvature give a positive invariant ball of radius 7.16×10^{-5} around the cap-zero minimum. Within this ball,

$$\|p(t) - p_*\|_2 \leq e^{-0.312161527688847t} \|p(0) - p_*\|_2.$$

The flow, its time, preparation, and gain are premises. This is neither a Born-rule probability nor a physical relaxation rate.

6 ST409–ST412: source stop, rank-seven obstruction, and invariants

6.1 Gain source and rank-seven nontransfer

ST409 finds no new strict typed object that fixes the sign and magnitude of g with coupling semantics. Deriving strong consequences from a supplied gain cannot be reversed into a source theorem.

The rank-seven Fourier mediator also remains unresolved globally. Its effective row weights include four negative entries, with minimum $-0.17141297146\dots$, so the positive-weight water-filling theorem is inapplicable. Its certified local root has maximum probability about 0.903306, below the approximately 0.93236 threshold at which the cap-curvature method becomes positive. ST410 closes these two proof routes only; it does not refute globality of the rank-seven state.

6.2 First forced invariant relations

The selected even D_{12} invariant family contains six quadratic generators, 32 primitive quartic generators, and 117 primitive sextic generators. Its formal degree-eight products count as follows:

$$\binom{9}{4} = 126, \quad \binom{7}{2} 32 = 672, \quad \binom{33}{2} = 528, \quad 6 \cdot 117 = 702.$$

The total is 2,028. Molien averaging gives dimension 1,892 for the complete mean-zero degree-eight invariant space. Therefore rank-nullity proves

$$\dim \ker(\text{degree-eight product map}) \geq 2028 - 1892 = 136.$$

Because the declared degree-six candidate family had zero relations, degree eight is the first forced relation degree in this chosen even generator construction. An explicit minimal syzygy basis remains open.

6.3 Algebraic versus statistical identifiability

Exact modular work gives algebraic rank 365 for the sextic Reynolds basis under a generic vertex-resolving evaluation design. ST412 evaluates those 365 directions at 500 frozen mean-zero unit vectors and normalizes every design column. Numerical rank remains 365, but

$$\sigma_{\min} = 3.04179 \times 10^{-4}, \quad \kappa_2 \approx 1.50346 \times 10^4.$$

For supplied observation noise 10^{-3} , the worst least-squares coefficient standard deviation is about 3.29. Thus exact distinguishability and stable noisy recovery are not the same theorem.

7 ST413: an adapted Chernoff cone

The previous synthetic Chernoff transfer certificate used the coarse derivative bound 6.6 and produced a gap near 2.5×10^{-72} . For a binary emission probability $q(b) = e_1 + (e_0 - e_1)b$ and Hilbert coordinate $h = \log(b/(1 - b))$, direct maximization gives

$$\max_b \left| \frac{d}{dh} \log q(b) \right| = \frac{|\sqrt{e_0} - \sqrt{e_1}|}{\sqrt{e_0} + \sqrt{e_1}}.$$

For the frozen emissions this is at most $1/3$. On $s \in [0.45, 0.55]$, the adapted product-metric weight Lipschitz constant is therefore $11/60$, not 6.6 . With filter contraction $5/7$, the invariant cone slope becomes $77/120$. The resulting projective-diameter bound is 9.19769861 , and Birkhoff's formula yields

$$1 - \tanh(9.19769861/4) \geq 0.0199262915891.$$

This improves the explicit bound by 69.90 decimal orders. The earlier microscopic gap was principally a metric/bounding artefact, not evidence of intrinsically slow mixing. The entire result remains conditional on the synthetic HMM fixture.

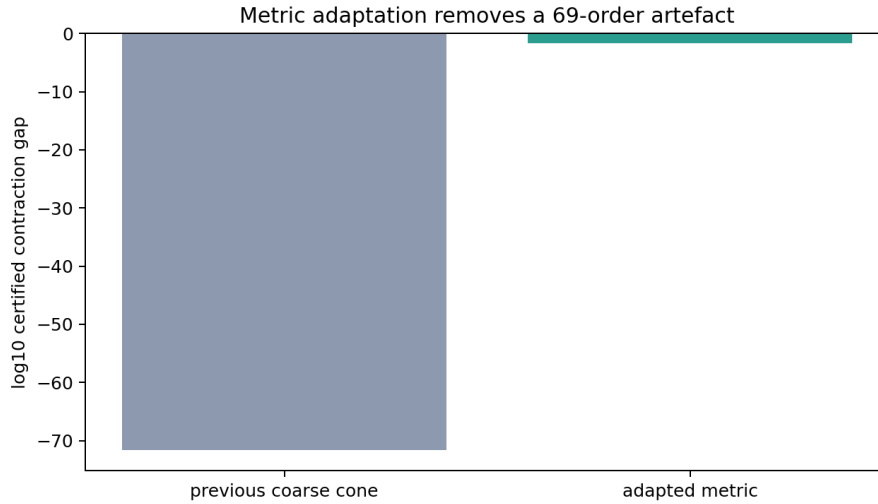


Figure 3: The adapted Hilbert-coordinate calculation removes the coarse-cone artefact.

8 ST414–ST415: expanded IR tube and a limiting band-loss face

8.1 A 55-fold wider root tube

The supplied continuum-IR KKT system has central root

$$(y_1, y_2, y_3, a, \nu) = (0.0281902, 2.1000678, 2.5590976, 0.0207319, 0.1077760) \dots$$

One common outward Krawczyk box now proves a unique root for every

$$B \in [2.9945, 3.0055], \quad T \in [3.9945, 4.0055],$$

using coordinate radii $(0.00055, 0.022, 0.0275, 0.00055, 0.0022)$. The minimum inclusion margin is 8.94×10^{-5} . This enlarges the ST395 parameter halfwidth by a factor 55. A fixed-template bisection accepts through $\delta = 0.00554468$ and first fails at 0.00554492 ; this brackets the limitation of that one box template, not the maximal mathematical continuation domain. The full complement-sign topology cover was not re-certified on this enlarged rectangle.

8.2 The compactified upper-band-loss face

Numerical continuation toward smaller T shows $a \downarrow 0$ and $y_3 \uparrow \infty$. Setting $a = 0$ and compactifying $y_3 = \infty$ leaves four equations for (y_1, y_2, ν, T) . A fresh four-dimensional Krawczyk calculation isolates their unique root in a radius- 10^{-8} box around

$$(0.028635829467935, 3.89120890375954, 0.0694500828220804, 2.44218197709456).$$

The inclusion margin is 9.9974×10^{-9} .

At this face the large- y linear term disappears. The third threshold escapes to infinity and the upper active band vanishes. What is proved is uniqueness of the compactified limiting-face root. The claim that the finite two-band branch attaches to it is supported by residual-controlled continuation, but not by an interval certificate for the singular attachment.

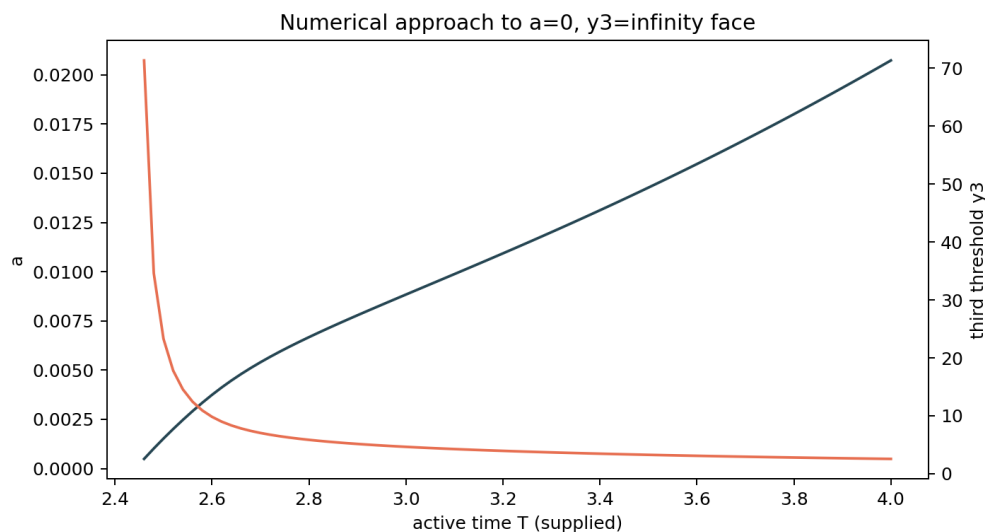


Figure 4: Numerical approach to the interval-certified compactified face. The displayed approach is evidence, not the face theorem itself.

9 Falsification ledger and scientific interpretation

The batch deliberately keeps negative outcomes visible.

- The exact orbit theorem does not select a realized vertex; it makes the selector obstruction sharper.
- The gain tube proves robustness under a supplied coefficient but provides no source for that coefficient.
- The branch-local crossing near 2.90249648 is not promoted to a global transition.
- The numerical Morse catalog is not an exhaustive stationary-point theorem.

- The rank-seven global problem survives; two imported proof mechanisms fail.
- Degree-eight rank-nullity proves a relation count lower bound, not explicit relations or a minimal presentation.
- The Chernoff improvement concerns a synthetic fixture, not a detector.
- ST414 certifies roots, not the full two-band topology on the enlarged box.
- ST415 certifies a limiting face, not its singular branch attachment and not a physical phase boundary.

The deepest new mathematical interpretation is therefore precise but limited: the supplied strict information functional has a robust, finite, symmetry-degenerate landscape whose complete global minimum set is now independently certified. It is a theorem about consequences of symmetry and information geometry. It is not yet a theorem explaining why Nature supplies the functional, the gain, an orbit member, or a dimensional realization.

10 Recommended next programs ST417–ST431

Priority	Program	Acceptance criterion
1	ST417 — explicit degree-eight syzygy basis	Construct and exactly verify at least 136 independent relations; separate a basis from a mere dimension count.
1	ST418 — global gain-transition enclosure	Combine gain-parametric cap/envelope proofs with certified competing branches; either isolate all global changes on a bounded range or report uncovered cells.
1	ST419 — singular IR attachment theorem	Blow up $a \rightarrow 0$, $y_3 \rightarrow \infty$ and interval-certify attachment to the ST415 compactified face.
2	ST420 — gain-tube enlargement	Optimize cap threshold and rational benchmarks to maximize the certified interval around $g = 4$; report template versus theorem maximality.
2	ST421 — stationary-orbit interval atlas	Certify the nine ST407 orbit candidates one by one and use a genuine global cover before claiming completeness.
2	ST422 — Morse barrier graph	Once stationary orbits are certified, interval-bound heteroclinic barrier values and adjacency without assigning physical transition rates.
2	ST423 — full enlarged IR topology cover	Restore complement-sign, inactive-slack, and tail certificates throughout the ST414 rectangle.
3	ST424 — invariant-ring presentation	Compute generators and relations jointly through degree eight, including odd-degree sectors; stop under a declared resource budget.
3	ST425 — optimal noisy invariant design	Choose evaluation states to maximize the smallest singular value under a frozen preparation/instrument class.
3	ST426 — Chernoff gap sharpness	Compare the adapted cone bound with direct transfer-operator discretization and obtain certified upper and lower contraction brackets.
3	ST427 — nonlinear local flow robustness	Add bounded perturbations to the declared projected flow and certify input-to-state stability of the local basin.

Priority	Program	Acceptance criterion
4	ST428 — rank-seven sign-aware envelope	Develop an envelope that permits sign-changing effective weights; success requires a rigorous bound overlapping the local-root region.
4	ST429 — source-admission theorem	Run only if a genuinely new strict typed object couples to g with sign and magnitude; consequence-to-source inversion is forbidden.
5	ST430 — selector/source theorem	Run only with a new non-premise symmetry-breaking provider; no replay of $\text{Aut}(\mathbb{Z}_{12})$ -invariant selectors.
5	ST431 — independent evidence gate	Require independently registered raw data or external audit; local code cannot satisfy this program.

The preferred local order is $\text{ST417} \rightarrow \text{ST419} \rightarrow \text{ST418}$, with ST423 and ST426 as independent secondary tracks. ST429–ST431 are evidence/source gated and must not be simulated by additional local algebra.

11 Reproducibility

The executable source is `fin_st402_st416_research.py`. It writes one JSON packet per program, the aggregate `FIN_ST402_ST416_Results.json`, a CSV ledger, and four figures. The suite `test_fin_st402_st416.py` contains 18 tests covering matrix structure, independent interval inclusion, outside separation, curvature, gain persistence, status boundaries, invariant counts, Chernoff constants, IR certificates, and packet hashes. All 18 tests pass in the recorded run. A SHA-256 manifest accompanies the release.

The fixed random seed is 20260822. The numerical stationary catalog and branch continuation are explicitly separated from interval-certified statements. Floating solver success is nowhere used as a theorem of global exhaustion or phase change.

References

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